



$$-t$$

$$|n-m| > 1 \Rightarrow \text{hopping} \approx 0$$

$$\int dt \psi_n^\dagger(r) V(r) \psi_m(r)$$

$$\langle n | \sum_m V | n \rangle = \begin{cases} V_0 & \text{if } n=m \\ -t & \text{if } |n-m|=1 \\ 0 & \text{OTHERWISE} \end{cases}$$

N nuclei

H is $N \times N$ matrix

Ansatz

$$\phi_n = \frac{e^{-ikna}}{\sqrt{N}}$$

$$k \leftrightarrow k + \left(\frac{2\pi}{a}\right)$$

$$\begin{aligned} \epsilon_0 &= -t \left(e^{-ik(n+1)a} + e^{-ik(n-1)a} \right) \\ &= E_0 e^{-ikna} \end{aligned}$$

$$\langle n | \sum_{\mathbf{r}} V(\mathbf{r}) | n \rangle = \begin{cases} V_0 & \text{if } n=0 \\ -t & \text{if } n=\pm 1 \\ 0 & \text{OTHERWISE} \end{cases}$$



EIGENSTATE ARE GIVEN

COUNTING STATES

WAVELET

$k = 0$

$k = \pi$

$$\frac{\partial \psi(x)}{\partial x} = \frac{1}{N} \sum_{\mathbf{k}} \psi(\mathbf{k})$$

$$N = 100$$

ENERGY BAND $\Rightarrow$

1 EIGENSTATE AT
EACH k IN THE
BRILLOUIN ZONE.

